Aravind Karthik R

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Education

- **Vellore Institute of Technology** — Chennai, India
  M.Tech – Embedded Systems — RTOS, MCU Architecture, ML/DL, In-Vehicle Networking — *Jul 2025 – Present*
- **Vellore Institute of Technology** — Chennai, India
  B.Tech – Electronics & Computer Engineering, Minor in Management — *Jul 2021 – Sep 2025*

Skills

- **Languages & HDL**: Embedded C, C++, Python, ARM Assembly, Verilog, CUDA
- **Embedded & OS**: RTOS (QNX), Embedded Linux, Linux Device Drivers, Firmware, Hardware Bring-Up, STM32, PSoC, NVIDIA Jetson
- **AI & Signal Processing**: TensorFlow Lite, PyTorch, OpenCV, scikit-image, Model Quantization, FFT, EEG/fNIRS
- **Protocols & Tools**: CAN, SPI, I2C, BLE, LoRa, MQTT, TCP/IP | Icarus Verilog, GTKWave, Quartus Prime, MATLAB

Professional Experience

- **Indian Institute of Science (IISc)** — Bangalore, India
  Project Associate Intern — *Jun 2024 – Jun 2025*
  - Designed and deployed real-time edge AI inference pipelines on Raspberry Pi, optimizing models for latency and memory under embedded constraints; demonstrated systems to 500+ visitors at IISc Open Day 2025.
  - Built and validated a biomedical dataset from 150+ participants; systematic augmentation yielded a **12% gain in vein detection accuracy** across end-to-end acquisition, training, quantization, and deployment workflow.
- **Stemtec AI and Robotics Pvt. Ltd. (V-NEST, VIT Chennai)** — Chennai, India
  Embedded Systems and IoT Intern — *Mar 2024 – Jun 2024*
  - Developed C/C++ firmware for measurement and control modules; performed PCB validation, hardware bring-up, and SMD soldering; integrated BLE and LoRa into low-power IoT communication systems.
- **Highbrow Diligence Services Pvt. Ltd.** — Chennai, India
  Embedded Systems Intern — *Aug 2023 – Dec 2023*
  - Developed an embedded automation prototype for a medical EBOO therapy device with multi-sensor SPI/I2C acquisition and safety-triggered actuator control.

Projects

- **Mitochondrial Activity Score (MAS) Pipeline** — M.Tech Thesis
  Python, scikit-learn, scikit-image, OpenCV, 3D-SIM Fluorescence Microscopy
  - Built a quantitative image analysis pipeline for 3D-SIM fluorescence microscopy (63×, 0.03 µm/px, 16-bit CZI); extracted 12 MiNA-grounded morphometric features via Sauvola thresholding, skeleton pruning, and EDT decomposition.
  - L1-regularized classifier achieved **AUC 0.891**, Cohen's $d = 1.80$ ($p = 4.74 \times 10^{-9}$), BH-FDR 5% significance on 9/12 features; quantified DRP1-mediated fragmentation: 33% lower fragment count, 25% smaller network area in patient cells.
- **Near-Infrared Vein Visualization System**
  DeepLabV3+, MobileNet, Raspberry Pi, RTX 3050
  - Non-contact 820 nm NIR imaging using hemoglobin differential absorption; DeepLabV3+ (MobileNet) trained on 150+ participants achieved **12% accuracy gain**; distributed Pi-to-GPU pipeline delivered real-time inference at **23 FPS**.
- **fNIRS Stroke Prediction & IV Monitoring Systems**
  ESP32, PSoC4, Raspberry Pi, AS7341, BLE, TCP
  - fNIRS: Dual-wavelength (780/820 nm) optical sensing with ESP32 ADC filtering, TCP-streamed to Raspberry Pi for ML stroke prediction; **end-to-end latency ∼250 ms**.
  - IV Monitor: Closed-loop capacitive fluid sensing with AS7341 spectral reverse-flow detection; motorized clamp actuation and BLE safety alerts, full pipeline on embedded hardware.
- **EEG-Based ASD Analysis**
  ESP32, MCP3204 12-bit ADC, Raspberry Pi
  - 8-channel EEG acquisition (frontal, temporal, occipital); extracted first-, second-, and third-order features; inter-channel covariance heatmaps quantified spectral coherence differences between ASD and TD groups.

Leadership & Volunteering

- **Programme Representative, M.Tech Embedded Systems – VIT Chennai**: Elected cohort representative; coordinated 6+ faculty and student body on scheduling and academic logistics.
- **Technical & Community Engagement**: Active member, Rotaract Club and Microsoft Innovation Club; contributed to technical workshops and peer learning.